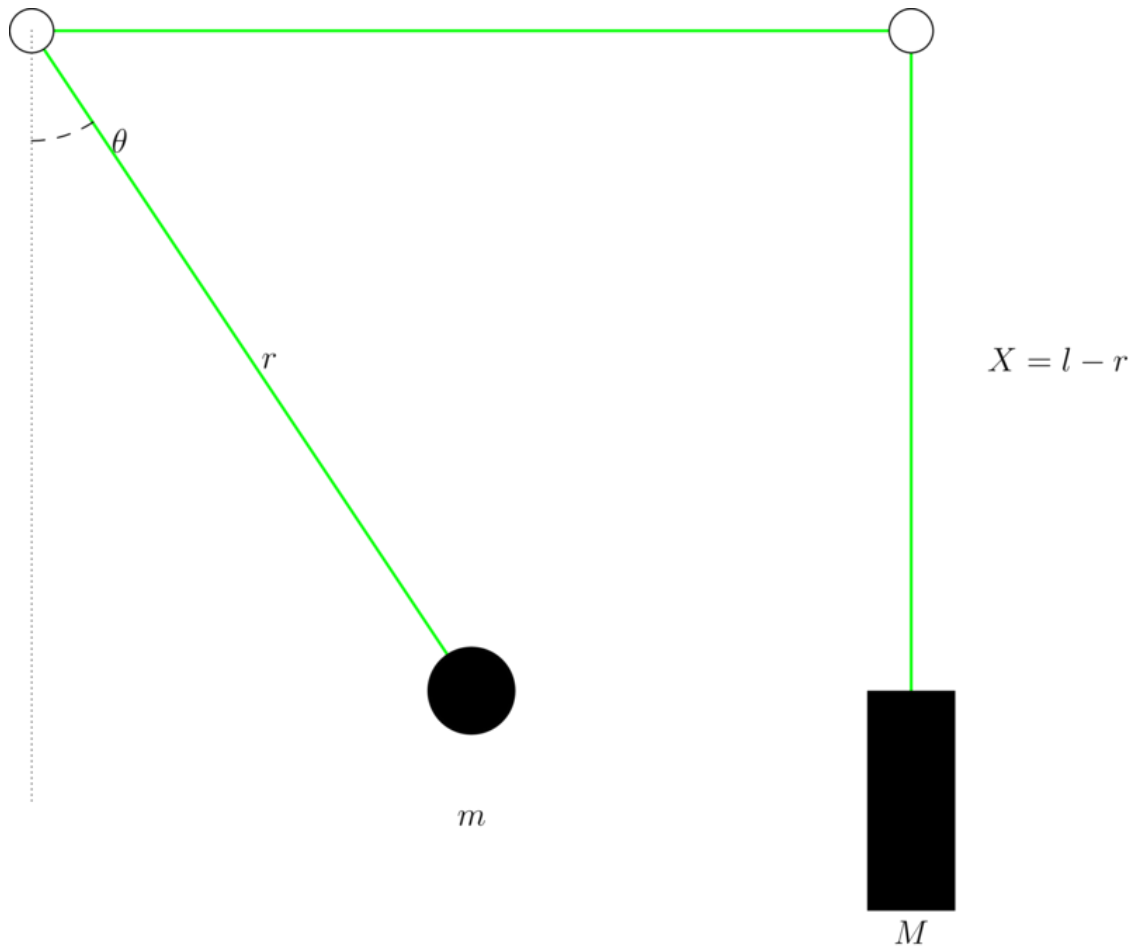


oscillating_atwoods_math

May 16, 2026

1 Oscilating Atwood Machine

Kartikeswar Rana



```
[1]: import sympy as sp
      sp.init_printing()
```

```
[2]: t,m1,m2,g,l = sp.symbols("t m_1 m_2 g l")
      r = sp.Function("r")(t)
      theta = sp.Function("theta")(t)
```

```
[3]: r_dot = sp.diff(r,t)
      theta_dot = sp.diff(theta,t)
      r_2dot = sp.diff(r_dot,t)
      theta_2dot = sp.diff(theta_dot,t)
```

1.1 Kinetic Energy

```
[4]: T = (1/2)*(m1+m2)*r_dot**2 + (1/2)*m1*r**2*theta_dot**2
      T
```

[4]:
$$0.5m_1r^2(t)\left(\frac{d}{dt}\theta(t)\right)^2 + (0.5m_1 + 0.5m_2)\left(\frac{d}{dt}r(t)\right)^2$$

1.2 Potential Energy

```
[5]: V = m1*g*r*(1-sp.cos(theta)) - m2*g*(l-r)
      V
```

[5]:
$$gm_1(1 - \cos(\theta(t)))r(t) - gm_2(l - r(t))$$

1.3 Langrangian

```
[6]: # langrangian
      L = T - V
      L
```

[6]:
$$-gm_1(1 - \cos(\theta(t)))r(t) + gm_2(l - r(t)) + 0.5m_1r^2(t)\left(\frac{d}{dt}\theta(t)\right)^2 + (0.5m_1 + 0.5m_2)\left(\frac{d}{dt}r(t)\right)^2$$

```
[7]: dLdrdot = sp.diff(L,r_dot)
      dLdrdotdt = sp.diff(dLdrdot,t)
      dLdr = sp.diff(L,r)
      EL_r = dLdrdotdt - dLdr
      EL_r
```

[7]:
$$gm_1(1 - \cos(\theta(t))) + gm_2 - 1.0m_1r(t)\left(\frac{d}{dt}\theta(t)\right)^2 + 2(0.5m_1 + 0.5m_2)\frac{d^2}{dt^2}r(t)$$

```
[8]: dLdthetadot = sp.diff(L,theta_dot)
      dLdthetadotdt = sp.diff(dLdthetadot,t)
      dLdtheta = sp.diff(L,theta)
      EL_theta = dLdthetadotdt - dLdtheta
      EL_theta
```

[8]:
$$gm_1r(t)\sin(\theta(t)) + 1.0m_1r^2(t)\frac{d^2}{dt^2}\theta(t) + 2.0m_1r(t)\frac{d}{dt}r(t)\frac{d}{dt}\theta(t)$$

1.4 Expression for $\ddot{r}$

[9]: `sp.solve(EL_r,r_2dot)`

[9]:
$$\left[\frac{gm_1 \cos(\theta(t)) - gm_1 - gm_2 + m_1 r(t) \left(\frac{d}{dt} \theta(t) \right)^2}{m_1 + m_2} \right]$$

1.5 Expression for $\ddot{\theta}$

[10]: `sp.solve(EL_theta,theta_2dot)`

[10]:
$$\left[-\frac{g \sin(\theta(t)) + 2.0 \frac{d}{dt} r(t) \frac{d}{dt} \theta(t)}{r(t)} \right]$$